

Rostand A. K. Fezeu

Curriculum Vitae

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Education

2021 - Present

Ph.D. Computer Science, University of Minnesota - Twin Cities.

Advisor. Zhi-Li Zhang, **Expected: Summer 2025**

2021

M.S. Computer Science, University of Minnesota - Twin Cities.

2017

B.S. Computer Science, Minnesota State University Moorhead.

B.S Mathematics, Emphasis in Computations, Minnesota State University Moorhead.

Publications

In Conference Proceedings and Workshops

- 2025 **[CCAT]** Rushikesh Zende, Sree Ganesh Lalitaditya Divakarla, Jason Carpenter, **Rostand A. K. Fezeu**, Ajay Kumar Gurumadiah, Eman Ramadan, and Zhi-Li Zhang. Poster: Impact of 5g mobility on teleoperation of cav. In *CCAT '25*. Global Symposium on Mobility Innovation, 2025.
- 2025 **[HotMobile]** **Rostand A. K. Fezeu**, Jason Carpenter, Eman Ramadan, Faaq Bilal, Ziyang Wu, Nanditha Naik, Duncan Joly, Dehkontee Chea Cuppah, and Zhi-Li Zhang. A peek into 5g nsa vs. sa control plane performance. In *Proceedings of the 26th International Workshop on Mobile Computing Systems and Applications*, HotMobile '25, page 49–54, New York, NY, USA, 2025. Association for Computing Machinery.
- 2025 **[MobiCom]** **UNDER REVIEW** **Rostand A. K. Fezeu**, Jason Carpenter, Faaq Bilal, Rushikesh Zende, Steven Sleder, Nitin Varyani, Nanditha Naik, Eman Ramadan, and Zhi-Li Zhang. Teleoperating connected and autonomous vehicles over commercial 5g networks: Are we there yet? In *The ACM Conference on Mobile Computing and Networking*, MobiCom '25, 2025.
- 2025 **[INFOCOM]** **UNDER REVIEW** **Rostand A. K. Fezeu**, Faaq Bilal, Claudio Fiandrino, Eman Ramadan, Dustin Maas, Joerg Widmer, and Zhi-Li Zhang. Performance implications of 5g tdd frame structure configuration of commercial 5g networks. In *IEEE International Conference on Computer Communications*, INFOCOM, 2025.
- 2025 **[SIGCOMM]** **UNDER REVIEW** Ahmad Hassan, Anlan Zhang, Jason Carpenter, **Rostand A. K. Fezeu**, Myungjin Lee, Ruiyang Zhu, Shuwei Jin, Wei Ye, Akshay Jajoo, Feng Qian, Morley Mao, and Zhi-Li Zhang. Opcm: Opportunistic performance-driven cell management for 5g/xg networks. In *Proceedings of the ACM*, SIGCOMM, 2025.
- 2024 **[SIGCOMM]** Wei Ye, Xinyue Hu, Steven Sleder, Anlan Zhang, Udhaya Kumar Dayalan, Ahmad Hassan, **Rostand A. K. Fezeu**, Akshay Jajoo, Myungjin Lee, Eman Ramadan, Feng Qian, and Zhi-Li Zhang. Dissecting carrier aggregation in 5g networks: Measurement, qoe implications and prediction. In *Proceedings of the ACM*, SIGCOMM, 2024.
- 2024 **[CCAT]** Nitin Varyani, Jason Carpenter, Steven Sleder, Wei Ye, **Rostand A. K. Fezeu**, Eman Ramadan, and Zhi-Li Zhang. Poster: Impact of data compression on ai perception tasks for teleoperated autonomous vehicles. In *CCAT '24*. Global Symposium on Mobility Innovation, 2024.
- 2024 **[CCAT]** Nitin Varyani, Jason Carpenter, Steven Sleder, Wei Ye, **Rostand A. K. Fezeu**, Eman Ramadan, and Zhi-Li Zhang. Poster: Cross-layer feasibility study of av teleoperations on 5g networks. In *CCAT '24*. Global Symposium on Mobility Innovation, 2024.
- 2024 **[SIGCOMM]** **Rostand A. K. Fezeu**, Claudio Fiandrino, Eman Ramadan, Jason Carpenter, Lilian Coelho de Freitas, Faaq Bilal, Wei Ye, Joerg Widmer, Feng Qian, and Zhi-Li Zhang. Unveiling the 5g mid-band landscape: From network deployment to performance and application qoe. In *Proceedings of the ACM*, SIGCOMM, 2024.

- 2024 **[INFOCOM]** **Rostand A. K. Fezeu**, Claudio Fiandrino, Eman Ramadan, Jason Carpenter, Daqing Chen, Yiling Tan, Joerg Widmer, Feng Qian, and Zhi-Li Zhang. Roaming across the european union in the 5g era: Performance, challenges and opportunities. In *IEEE International Conference on Computer Communications, INFOCOM*, 2024.
- 2023 **[IEEE MeditCom]** Wei Ye, Jason Carpenter, Zejun Zhang, **Rostand A. K. Fezeu**, Feng Qian, and Zhi-Li Zhang. A closer look at stand-alone 5g deployment from the ue perspective. In *IEEE MeditCom 2023: IEEE International Mediterranean Conference on Communications and Networking 2023*, IEEE MeditCom 2023. Association for Computing Machinery, 2023.
- 2023 **[arXiv]** **Rostand A. K. Fezeu**, Timothy J Salo, Amy Zhang, and Zhi-Li Zhang. Dissecting iot device provisioning process. *arXiv preprint arXiv:2310.14125*, 2023.
- 2023 **[PAM]** **Rostand A. K. Fezeu**, Eman Ramadan, Wei Ye, Benjamin Minneci, Jack Xie, Arvind Narayanan, Ahmad Hassan, Feng Qian, Zhi-Li Zhang, Jaideep Chandrashekar, et al. An In-Depth measurement analysis of 5G mmwave PHY latency and its impact on End-to-End delay. In *International Conference on Passive and Active Network Measurement, PAM '23*, pages 284–312. Springer, Springer Nature Switzerland, 2023.
- 2023 **[arXiv]** **Rostand A. K. Fezeu**, Jason Carpenter, Claudio Fiandrino, Eman Ramadan, Wei Ye, Joerg Widmer, Feng Qian, and Zhi-Li Zhang. Mid-band 5g: A measurement study in europe and us. In *arXiv*. arXiv, 2023.
- 2023 **[IEEE NETWORK]** Udhaya Kumar Dayalan, Timothy J. Salo, **Rostand A. K. Fezeu**, and Zhi-Li Zhang. Kaala 2.0: Scalable iot/nextg system simulator. *IEEE Network*, volume 37, pages 240–246, 2023.
- 2022 **[IEEE DCOSS]** Udhaya Kumar Dayalan, **Rostand A. K. Fezeu**, Timothy J. Salo, and Zhi-Li Zhang. Prototyping a fine-grained qos framework for 5g and nextg networks using powder. In *2022 18th International Conference on Distributed Computing in Sensor Systems (DCOSS)*, pages 416–419, 2022.
- 2022 **[SIGCOMM NET4us]** Udhaya Kumar Dayalan, **Rostand A. K. Fezeu**, Timothy J. Salo, and Zhi-Li Zhang. Kaala: Scalable, end-to-end, iot system simulator. In *Proceedings of the ACM SIGCOMM Workshop on Networked Sensing Systems for a Sustainable Society, NET4us '22*, page 33–38, New York, NY, USA, 2022. Association for Computing Machinery.
- 2021 **[IMC POSTER]** **Rostand A. K. Fezeu**, Timothy J Salo, Amy Zhang, and Zhi-Li Zhang. Poster: Unveiling iot devices provisioning process. In *Proceedings of the ACM Internet Measurement Conference, IMC '21*. Association for Computing Machinery, 2021.
- 2021 **[SIGCOMM 5G-MeMU]** Eman Ramadan, Arvind Narayanan, Udhaya Kumar Dayalan, **Rostand A. K. Fezeu**, Feng Qian, and Zhi-Li Zhang. Case for 5g-aware video streaming applications. In *Proceedings of the 1st Workshop on 5G Measurements, Modeling, and Use Cases, 5G-MeMU '21*, page 27–34, New York, NY, USA, 2021. Association for Computing Machinery.
- 2021 **[IEEE CCGrid]** Udhaya Kumar Dayalan, **Rostand A. K. Fezeu**, Nitin Varyani, Timothy J. Salo, and Zhi-Li Zhang. Veeredge: Towards an edge-centric iot gateway. In *2021 IEEE/ACM 21st International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, pages 690–695, 2021.
- 2021 **[HotMobile]** Udhaya Kumar Dayalan, **Rostand A. K. Fezeu**, Nitin Varyani, Timothy J. Salo, and Zhi-Li Zhang. Eciot: Case for an edge-centric iot gateway. In *Proceedings of the 22nd International Workshop on Mobile Computing Systems and Applications, HotMobile '21*, page 154–156, New York, NY, USA, 2021. Association for Computing Machinery.
- 2020 **[ICNP]** **Rostand A. K. Fezeu** and Zhi-Li Zhang. Anomalous model-driven-telemetry network-stream bgp detection. In *2020 IEEE 28th International Conference on Network Protocols (ICNP)*, pages 1–6, 2020.
- 2020 **[IMC]** Arvind Narayanan, Eman Ramadan, Rishabh Mehta, Xinyue Hu, Qingxu Liu, **Rostand A. K. Fezeu**, Udhaya Kumar Dayalan, Saurabh Verma, Peiqi Ji, Tao Li, Feng Qian, and Zhi-Li Zhang. Lumos5g: Mapping and predicting commercial mmwave 5g throughput. In *Proceedings of the ACM Internet Measurement Conference, IMC '20*, page 176–193, New York, NY, USA, 2020. Association for Computing Machinery.

Research Experience

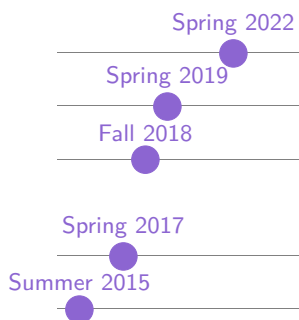
May. - Aug.2024	FutureWei Technologies, Dallas, Texas, USA <i>Wireless Systems Architecture for 5G & 6G, Research Intern.</i> Worked on Network Assisted Rate Control for Immersive 5G and 6G Applications
Oct. - Dec.2022	IMDEA Networks Institute, Madrid, Spain <i>5G Networks Visiting Researcher.</i> Orchestrated and conducted the largest cross border 5G Mid-Band data collection campaign in Europe.
May - Aug.2022	Microsoft, Redmond, Washington, USA <i>PhD Software Engineering Intern, Research Project.</i> Build a framework to test the scalability of SdnExpress, a Software Defined Network SaaS framework.
May - Aug.2021	<i>PhD Software Engineering Intern, Research Project.</i> Design and prototyped FleetAgent: A large scale network performance monitoring tool.
Jan.2014 - May.2017	Minnesota State University Moorhead, MSUM, Moorhead, Minnesota, USA <i>A Possible Generalization of Bertrand's Postulate, Undergraduate Research.</i> Studied Bertrand's Postulate and investigated the conjecture: Between two consecutive multiples of a positive interger, n where $n \geq 2$, there is always an integer, m divisible by an odd prime number, p greater than n
Jan.2014 - May.2017	<i>Radio Frequency Identification (RFID).</i> Analyzed ethical issues associated with RFIDs implantation in human bodies including Violotion of Privacy, Security Concerns, RFID usage in Medical field and supply chaning management systems.

Related Employment

Aug.2018 - Present	University of Minnesota – Twin Cities, Minneapolis, USA <i>Research Assistant.</i> Working on a variety of projects including Internet of Things (IoT), Software Defined Networking (SDN), 5G Networks, Network Automation and Control, etc.
May - Aug.2022	Microsoft, Redmond, Washington, USA <i>Software Engineering, PhD Intern.</i> Build a framework to test the scalability of SdnExpress, a Software Defined Network SaaS framework.
May - Aug.2021	<i>Software Engineering, PhD Intern.</i> Design and prototyped FleetAgent: A large scale network performance monitoring tool.
July - June.2020	Cisco Systems, San Jose, California, USA <i>Data Scientist, PhD Intern.</i> Build a data preprossing pipeline for detecting anomalies in streaming telemetry data using unsupervised Machine Learning.
Aug.2017 - Aug.2018	Information Mangement Systems, IMS, Fargo, North Dakota, USA <i>Network Administrator/Consultant.</i> Installed, deployed, configured and managed network devices including Citrix, Hyper-V and VMware servers, Microsoft active directory, VoIP, etc.
May.2014 - May.2017	Minnesota State University Moorhead, Moorhead, Minnesota, USA <i>Information Technology Tier I and Tier II Support.</i> Diagnosed and troubleshot network connectivity issues including telephony, printers, computers, etc.
2013 - March2017	<i>Janitorial Worker, SODEXO Campus Dining.</i> Provided janitorial services on the campus dining center including, vacuuming, floor care, trash removal for over 5000+ students.

Teaching Experience/Assistantship

Spring 2022	University of Minnesota – Twin Cities, Minneapolis, Minnesota, USA CSCI 8211: Advanced Computer Networks (Graduate Course).
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CSCI 4211: Introduction Computer Networks.

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Minnesota State University Moorhead, Moorhead, Minnesota, USA

MATH 366: Differential Equations.

MATH 626: Calculus II.

Academic Achievements & Awards

\$800 HotMobile 25 Travel Grant.

\$1000 Global Symposium on Mobility Innovation Student Poster Competition Graduate Runner-up Awards.

\$1000 SIGCOMM 2023 Travel Grant.

\$1500 CoNEXT 2022 Travel Grant.

One of 12 students selected out of 57 applicants at the 3rd Annual AT&T Labs Graduate Student Symposium.

3rd Annual AT&T Labs Graduate Student Symposium Travel Grant. \$750.

Alumni Foundation Math Scholarship \$1000.

Alumni Foundation Computer Science Scholarship \$500.

8th at the Digi-Key Programming Competition.

Sophomore Mathematics Awards of \$500.

Sophomore Mathematics Awards of \$500.

2st at the second MSUM Annual Math Contest.

1st in Minnesota State University Moorhead MAA-NCS team contest.

14st out of 100 teams at eh MAA-NCS team contest.

Certifications and Continuing Education Units (CEUs).

IEEE Machine Type Communication in 5G and Beyond.

IEEE 5G Security.

IEEE 5G Wireless Positioning and Location.

IEEE Next-Generation Multi-Access Service Based 5G Architecture.

Cybersecurity Skills Assessment (CSI) boot camp.

Sophos firewall Certified Engineer.

Presentations

Examining Challenges Governing Self-Driving/Autonomous Networks, 3rd Annual AT&T Labs Graduate Student Symposium.

Cryptographic Properties of Hash Functions to Ensure Maximum Security, Minnesota State University Moorhead Student Academic Conference.

Feb. 2 2016

A Possible Generalization of Bertrand's Postulate, *Minnesota State University Moorhead Student Academic Conference twice., 37th Pi Mu Epsilon Conference College. College of St. Benedict/St. John's University, Minnesota State University Moorhead Mathematic Seminar.*

Dec. 2016

RSA Encryption, *Minnesota State University Moorhead Mathematics Seminar.*

Affiliations

- **National Association of Black Accountants (NABA), Student Member.**
- **Black Graduate and Professional Student Association (bgapsa) UMN..**
- **Golden Key International Honor Society, Student Member.**
- **Institute of Electrical and Electronics Engineers (IEEE), Student Member.**
- **Mathematical Association of America (MAA), Student Member.**
- **Minnesotat State University Moorhead Table Tennis Club, Founder and president.**
- **Chi-Alpha Christian Ministry, Small Grup Leader at MSUM.**
- **Minnesota State University Moorhead Mathematics Club.**

Computer skills

Research Interests	5G, Machine Learning, Systems and Networking – SDN, NFC, 5G RAN & Core Network, IoT, Smart Homes, Self-Driving Networks, Security and Privacy, 3GPP LTE and 5G, MIMO, 5G PHY Layer (L1), 5G MAC Layer (L2), Teleoperations, Autonomous Driving, Teleoperated Driving
Tools	srsRAN, OpenAIR, O-RAN, XCAL/XCAP, QXDM, DM, adb, diag, ROS, CARLA, GStreamer
Programming Languages	Python, C/C++, racket, Maple, LaTeX, MS Office, HTML, CSS, JavaScript, Bach Scripting, Linux commands, PowerShell scripting, Machine Learning, Data Mining,
Operating Systems	Linux, Mac OS, Windows
Database	MySQL, SQL, Mongo db, Redis
Language	Fluent in English and French.

Hobbies

- **Running**, *Ran 310 miles summer 2019, 2020, 2021 and 2022 – 10k 4 times a week.*
- **Biking, Swimming, Playing/Watching Soccer, Travelling.**

References

Dr. Zhi-Li Zhang

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University of Minnesota – Twin Cities
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Dr. Claudio Fiandrino

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IMDEA Networks Institute
Spain, Madrid
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Dr. Feng Qian

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Computer Science & Engineering
University of Minnesota – Twin Cities
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Dr. Joerg WIDMER

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IMDEA Networks Institute
Spain, Madrid
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Dr. Jaideep Chandrashekar

Principal Scientist
Interdigital Labs
Palo Alto, California, USA
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